

Bence bu projeyi **gerçek bir Data Analyst portföy projesi** gibi hazırlayalım. Amacımız sadece ödevi teslim etmek değil; Tableau Public'de paylaşabileceğin ve CV'ne koyabileceğin bir çalışma çıkarmak.

Ben olsam veri modelini şu şekilde kurarım:

```
BigQuery
|
├─ Query 1 → KPI
├─ Query 2 → Funnel
├─ Query 3 → Traffic Sources
├─ Query 4 → Device Analysis
├─ Query 5 → Landing Page Analysis
└─ Query 6 → Daily Performance

↓

Tableau Dashboard
```

Query 1 — KPI (Dashboard'ın üst kısmındaki kartlar)

Bu sorgu Tableau'da KPI kartlarını besleyecek.

```
<> SQL

WITH events AS (

SELECT

event_name,

CONCAT(
user_pseudo_id,
'-' ,
(
SELECT value.int_value
FROM UNNEST(event_params)
WHERE key='ga_session_id'
)
) AS session_id,

user_pseudo_id,

ecommerce.transaction_id,

ecommerce.purchase_revenue_in_usd AS revenue
```

```

FROM `bigquery-public-data.ga4_obfuscated_sample_ecommerce.events_*`

)

SELECT

COUNT(DISTINCT CASE
WHEN event_name='session_start'
THEN session_id
END) AS visits,

COUNT(DISTINCT user_pseudo_id) AS users,

COUNT(DISTINCT CASE
WHEN event_name='purchase'
THEN transaction_id
END) AS orders,

ROUND(SUM(CASE
WHEN event_name='purchase'
THEN revenue
ELSE 0
END),2) AS sales,

ROUND(
COUNT(DISTINCT CASE
WHEN event_name='purchase'
THEN transaction_id
END)
/
COUNT(DISTINCT CASE
WHEN event_name='session_start'
THEN session_id
END)
*100
,2) AS conversion_rate,

ROUND(

SUM(CASE
WHEN event_name='purchase'
THEN revenue
ELSE 0
END)

/

COUNT(DISTINCT CASE
WHEN event_name='purchase'
THEN transaction_id
END)

```

```
,2) AS average_order_value
```

```
FROM events;
```

Query 2 — Funnel

SQL



```
WITH events AS (  
  
SELECT  
  
event_name,  
  
CONCAT(  
user_pseudo_id,  
'-',  
(  
SELECT value.int_value  
FROM UNNEST(event_params)  
WHERE key='ga_session_id'  
)  
) AS session_id  
  
FROM `bigquery-public-data.ga4_obfuscated_sample_ecommerce.events_*`  
  
)  
  
SELECT  
  
COUNT(DISTINCT CASE WHEN event_name='session_start' THEN session_id END) AS session_start,  
  
COUNT(DISTINCT CASE WHEN event_name='view_item' THEN session_id END) AS view_item,  
  
COUNT(DISTINCT CASE WHEN event_name='add_to_cart' THEN session_id END) AS add_to_cart,  
  
COUNT(DISTINCT CASE WHEN event_name='begin_checkout' THEN session_id END) AS begin_checkout,  
  
COUNT(DISTINCT CASE WHEN event_name='add_shipping_info' THEN session_id END) AS add_shipping_info,  
  
COUNT(DISTINCT CASE WHEN event_name='add_payment_info' THEN session_id END) AS add_payment_info,  
  
COUNT(DISTINCT CASE WHEN event_name='purchase' THEN session_id END) AS purchase  
  
FROM events;
```

Bu sorgudan Tableau'da Funnel Chart oluşturacağız.

Query 3 — Traffic Sources

Bu grafik pazarlama ekibi için en önemlisi olacak.

</> SQL



```
WITH events AS (  
  
SELECT  
  
event_date,  
  
traffic_source.source,  
  
traffic_source.medium,  
  
traffic_source.name AS campaign,  
  
CONCAT(  
user_pseudo_id,  
'-',  
(  
SELECT value.int_value  
FROM UNNEST(event_params)  
WHERE key='ga_session_id'  
)  
) AS session_id,  
  
event_name,  
  
ecommerce.purchase_revenue_in_usd revenue  
  
FROM `bigquery-public-data.ga4_obfuscated_sample_ecommerce.events_*`  
  
)  
  
SELECT  
  
source,  
  
medium,  
  
campaign,  
  
COUNT(DISTINCT CASE WHEN event_name='session_start' THEN session_id END) visits,  
  
COUNT(DISTINCT CASE WHEN event_name='purchase' THEN session_id END) purchases,  
  
ROUND(SUM(CASE WHEN event_name='purchase' THEN revenue END),2) sales  
  
FROM events
```

```
GROUP BY
source,
medium,
campaign

ORDER BY sales DESC;
```

Query 4 — Device Analysis

</> SQL



```
WITH events AS (

SELECT

device.category,

device.language,

device.operating_system,

event_name,

CONCAT(
user_pseudo_id,
'-' ,
(
SELECT value.int_value
FROM UNNEST(event_params)
WHERE key='ga_session_id'
)
) AS session_id,

ecommerce.purchase_revenue_in_usd revenue

FROM `bigquery-public-data.ga4_obfuscated_sample_ecommerce.events_*`

)

SELECT

category,

language,

operating_system,

COUNT(DISTINCT CASE WHEN event_name='session_start' THEN session_id END) visits,
```

```
COUNT(DISTINCT CASE WHEN event_name='purchase' THEN session_id END) purchases,

ROUND(SUM(CASE WHEN event_name='purchase' THEN revenue END),2) sales

FROM events

GROUP BY
category,
language,
operating_system;
```

Query 5 — Landing Pages

Landing page, proje dokümanında özellikle isteniyor. Bunun için `page_location` veya `page_path` bilgisini `event_params` içinden çekmemiz gerekir.

<> SQL



```
WITH events AS (

SELECT

(
SELECT value.string_value
FROM UNNEST(event_params)
WHERE key='page_location'
) AS landing_page,

event_name,

CONCAT(
user_pseudo_id,
'-' ,
(
SELECT value.int_value
FROM UNNEST(event_params)
WHERE key='ga_session_id'
)
) session_id,

ecommerce.purchase_revenue_in_usd revenue

FROM `bigquery-public-data.ga4_obfuscated_sample_ecommerce.events_*`

)

SELECT
```

```
landing_page,

COUNT(DISTINCT CASE WHEN event_name='session_start' THEN session_id END) visits,

COUNT(DISTINCT CASE WHEN event_name='purchase' THEN session_id END) purchases,

ROUND(SUM(CASE WHEN event_name='purchase' THEN revenue END),2) sales

FROM events

GROUP BY landing_page

ORDER BY visits DESC;
```

Query 6 — Daily Performance

Bu sorgu zaman serisi (Time Series) grafiğini oluşturacak.

SQL



```
WITH events AS (

SELECT

PARSE_DATE('%Y%m%d',event_date) AS date,

event_name,

CONCAT(
user_pseudo_id,
'- ',
(
SELECT value.int_value
FROM UNNEST(event_params)
WHERE key='ga_session_id'
)
) session_id,

ecommerce.transaction_id,

ecommerce.purchase_revenue_in_usd revenue

FROM `bigquery-public-data.ga4_obfuscated_sample_ecommerce.events_*`

)

SELECT
```

```
date,  
  
COUNT(DISTINCT CASE WHEN event_name='session_start' THEN session_id END) visits,  
  
COUNT(DISTINCT CASE WHEN event_name='purchase' THEN transaction_id END) orders,  
  
ROUND(SUM(CASE WHEN event_name='purchase' THEN revenue END),2) sales  
  
FROM events  
  
GROUP BY date  
  
ORDER BY date;
```

Tableau Dashboard Taslağı

Bu 6 sorguyla aşağıdaki dashboard'u oluşturabiliriz:

Visits | Users | Orders | Sales | CR | AOV

Funnel (7 Adım)

Sales by Source Visits by Device

Landing Pages Daily Revenue Trend

Filters

✓ Date

✓ Source

✓ Medium

✓ Campaign

✓ Device Category

✓ Language

✓ Operating System

✓ Landing Page

Bir önemli öneri

Bu sorgular çalışacaktır, ancak ben bunları daha da profesyonel hale getirmek isterim.

Özellikle aynı `session_id` hesaplamasını ve `UNNEST` işlemlerini her sorguda tekrar etmek

yerine, önce **tek bir** `session_events` **tablosu (veya view)** oluřturup diğeri 6 sorguyu onun üzerinden yazabiliriz. Bu yaklařım:

- Kod tekrarını azaltır.
- Sorguları daha hızlı okunur hale getirir.
- Gerçek řirketlerde kullanılan veri modeline daha yakındır.

Ben bu ikinci yaklařımı tavsiye ederim; hem proje deęerlendirmesinde hem de portföyünde çok daha güçlü görünür.